DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

An Internship Report
On

"ICT & Visual Inspection"

A report submitted to Dr. Babasaheb Ambedkar Technological University in partial fulfilment of requirement for the award of the degree

BACHELOR OF ENGINEERING
IN
ELECTRONICS AND TELECOMMUNICATION ENGINEERING

Submitted by

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(PRN NO. 2221311372017)

Under the Guidance of

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DEPARTMENT OF ELECTRONICS AND TELECOMMUNICATION ENGINEERING

CERTIFICATE

This is to certify that the internship work on **"ICT & Visual Inspection"** is a bonafide work carried out by **Mr. Sk Albksh Sk Afzal (PRN NO. 2221311372017)** in partial fulfilment for the award of degree of **Bachelor of Technology** in the Department of Electronics and Telecommunication Engineering, offered by **Dr. Babasaheb Ambedkar Technological University (DBATU), LONERE,** during the academic year 2024-2025. It is certified that all corrections/suggestion indicated for the external assessment have been incorporated in the report deposited in the departmental library. The internship report has been approved as it satisfies the academic requirements prescribed for the Bachelor of Engineering degree.

Internship Guide
Mr. Suryakant Jadhav
Star Engineering, Pune

Internship Coordinator
Prof. S. L. Burange.

Head of department
Prof. V. V. Patange.

Principal
Dr. O. S. Rajankar.

DECLARATION

Mr. Sk Albksh Sk Afzal, a student of **Shri Tuljabhavani College Of Engineering Tuljapur, Maharashtra**, hereby declare that the internship report on **"ICT & Visual Inspection"** in **MI Department at Star Engineers India Pvt. Ltd., Chakan, Pune** submitted by me is a Bonafide record of the work carried out during my internship from **19th Feb 2026 to 19th May 2026** .

This report is submitted towards the **partial fulfilment of the requirements** for the degree of **Bachelor of Engineering in Electronics & Telecommunication Engineering** under **Dr. Babasaheb Ambedkar Technological University, Lonere**.

Ifurther declare that this work has **not been submitted anywhere else**, either in part or in full, for the award ofany other degree or diploma in this institute or any other university.

Place: Tuljapur
Date:

Name and Signature:

Mr. Sk Albksh Sk Afzal
(PRN NO. 2221311372017)

ACKNOWLEDGEMENT

With immense pleasure and great respect. I take this opportunity to express my deep sense of gratitude toward my guide, **Mr. Suryakant Jadhav,** Department supervisor at Star Engineering, Pune. for valuable guidance, inspiration, constant encouragement and motivation throughout the project work.

I am grateful to **Prof. V. V. Patange.**, Head of Department, Electronics and Telecommunication Engineering and the faculty of department Electronics and Telecommunication Engineering, **STBCE**, Tuljapur for providing the amicable environment and allowing me to useful available facilities in this department during the internship.

We thank to **Dr. O. S. Rajankar** our beloved principal, for his encouragement and providing all facilities needed in our project.

I am thankful to, **Prof. S. L. Burange** Internship Coordinator and all the respected, Professor from the department of Electronics and Telecommunication Engineering, STBCE, Tuljapur, who helped me to understand basic methodologies of my internship.

I am very thankful to my friend and well-wisher who directly or indirectly helped me at every stage to complete this work.

Last but not least, I would like to thanks my family and almighty without their good wishes and blessing this dissertation work could not have been completed.

Submitted by

Mr. Sk Albksh Sk Afzal
(PRN NO. 2221311372017)

INTERNSHIP LETTER

Date : 19/05 /2026

To,
Albaksh Afzal Shaikh

Subject: INTERNSHIP LETTER

Dear Albaksh Afzal Shaikh,

This is to certify that **Mr. Albaksh Afzal Shaikh** was employed with **Star Engineers India Pvt.ltd.** as a Visual Inspection & ICT Operator in MI Department from **19 Feb 2026 to 19 May 2026**. During this period, he successfully completed 03 Month of Internship. Throughout his tenure, she demonstrated commendable dedication, professionalism, and competence in his role.

he consistently performed her tasks with diligence, efficiency, and attention to detail, contributing significantly to the success of our team and the achievement of our organizational goals.

We wish him all the best in his future endeavors and have no hesitation in recommending her for any suitable opportunities.

Thank you for her dedicated service and contributions to **Star Engineers India Pvt.ltd**. We are grateful for her efforts and wish her continued success in her career.

TALENTCORP SOLUTIONS PRIVATE LIMITED PUNE

Sincerely,
HR Department
TALENTCORP SOLUTIONS PVT LTD

Corporate Office Address :
Talentcorp Solutions Pvt. Ltd.
First Floor, Shree Gajanan Commercial Complex, Chakan-Talegaon Road, Chakan Tal Khed, Dist. Pune, Maharashtra, India- 410501
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CHAPTER 1: INTRODUCTION

As part of the academic curriculum for the final year of Bachelor of Engineering in Electronics and Telecommunication, students are required to undergo an industrial internship. This internship is designed to bridge the gap between theoretical knowledge and real-world industrial practices.
I, Sk Albksh Sk Afzal, undertook my 3-month internship at **Star Engineers India Pvt. Ltd., Chakan, Pune**, a renowned electronics manufacturing company serving the automotive and non-automotive sectors.

The objective of this internship was to gain hands-on experience in electronics manufacturing, testing, and production support. I was assigned to the **MI Department (Production Section)**, where I worked as an **ICT Intern** from **19th** Feb 2026 to 19th May 2026. This department plays a vital role in ensuring the quality and efficiency of electronic component production.

During the course of my internship, I was involved in tasks such as **testing PCB components**, assisting in **production documentation**, and providing **technical production support**. These responsibilities helped me understand the practical implementation of engineering principles in a manufacturing environment.

Ihad the opportunity to work with various industry-standard tools and equipment such as **Kyoritsu Software**, **in-circuit testing fixtures**, and **PC-based diagnostic tools**. These tools were integral in performing detailed inspections and ensuring quality standards were maintained throughout the production line.

Working under the guidance of experienced professionals like **Mr. Suryakant Jadhav (Supervisor)** and mentors **Mr. Rahul Tarale** and **Mr. Datta Chavan**, I learned valuable industry practices and improved my technical and interpersonal skills. Their continuous support and knowledge sharing made the experience both educational and inspiring.

This internship helped me develop key professional skills such as **optical and visual inspection**, **problem-solving**, **plan and production management**, **safety evaluation**, and **effective communication and teamwork**. It also gave me a clearer understanding of the roles and responsibilities of an engineer in an actual industry setup.

The internship not only strengthened my technical foundation but also built my confidence in working within a structured industrial environment. Observing and being part of the process from component assembly to quality testing enhanced my understanding of real-world electronics manufacturing and automation.

Overall, this internship was a milestone experience that significantly contributed to my growth as an aspiring electronics engineer. It laid the groundwork for my future career path by helping me connect classroom knowledge with practical applications. This report summarizes the various aspects of my internship journey, including the company profile, tasks performed, tools used, skills developed, and the key learnings I acquired during the 3-month period.

CHAPTER 2: COMPANY OVERVIEW

Star Engineers India Pvt. Ltd.

Star Engineers India Pvt. Ltd., founded in **1988** by Mr. Kishorilal Ramraika, has grown into a global leader in electronic component manufacturing. Based in Pune with five manufacturing units—including two in Chakan—and an overseas facility in Vietnam, the company employs over **1,200** professionals and is equipped to produce upwards of **50 million parts per year**.

The company specializes in delivering high-quality electronic solutions to both **automotive and non-automotive sectors**. Its diverse product portfolio includes ignition controls, sensors, regulators & rectifiers, flashers, USB chargers, speedometers, battery chargers, motor-control units, converters and electronic throttles.

Star Engineers emphasizes **Quality, Cost & Delivery (QCD)** principles, and embraces continuous improvement through practices like **5S, TPM, Kaizen**, and stringent process control. The company holds certifications such as **TS16949** and **ISO 14000**, and holds more than **seven patents**, showing a forward-looking approach toward innovation—especially in the EV (electric vehicle) domain.

Organizational Structure & Leadership:

- **Chairman:** Mr.KishorilalRamraika
- **Managing Director:** Mr.DivyaRamraika
- **CEO:** Mr. Neeraj Kumar Malik

Corporate Social Responsibility (CSR): The company actively engages in CSR activities focused on environmental conservation and uplifting underprivileged communities, reflecting its commitment to sustainable and ethical business practices.

Global Footprint: Besides its core operations in Pune (Chakan and Talawade) and Aurangabad, Star Engineers has expanded overseas with a production facility in Vietnam to serve international markets.

Star Engineers India Pvt. Ltd., headquartered in Chakan, Pune, is a distinguished manufacturer specializing in electronic components for both automotive and non-automotive sectors. Established with the mission to deliver precision and innovation, the company has built a reputation for engineering excellence and commitment to quality across the industry.

Vision and Mission

Vision : Tobe a global leader in electronic component manufacturing by consistently delivering superior quality, innovation, and reliability.

Mission: To provide value-driven, technologically advanced products through continual investment in state-of-the-art infrastructure, skilled personnel, and R&D initiatives.

Core Values and Culture

StarEngineersfostersa **collaborative, proactive, and safety-focused** work environment that encourages **continuous learning** and open communication. Employees report high satisfaction with the culture—highlighting supportive management, transparent feedback processes, and a strong emphasis on professional growth and cross-functional teamwork.

Organizational Structure

The company operates under a functional organizational structure with distinct departments:

Department	Role & Function
R&D & Design	Product innovation, prototyping, and testing
MI (Production)	Electronics assembly, in-circuit testing, quality assurance
Quality Control	Ensuring compliance with national and international standards
Supply Chain & Logistics	Procurement, inventory, and delivery coordination
HR & Administration	Employee relations and development
Sales & Customer Support	Client interactions and after-sales services

Star Engineers also emphasizes **continuous improvement**, regularly conducting Kaizen and quality audits to optimize production processes and maintain ISO-compliant standards.

Facilities & Manufacturing Capability

Situated in Chakan's industrial hub, the company's **modern facility** includes:

- Advanced assembly lines for PCB fabrication
- In-circuit testing booths with **Kyoritsu Software**
- Optical inspection systems for surface-level checks
- Dedicated calibration and testing labs
- In-house R&D center for product enhancements

This infrastructure enables Star Engineers to deliver diverse products such as automotive sensors, controllers, diagnostic tools, and other specialized components with rigorous attention to quality.

Workforce & Talent Development

With a workforce of approximately 300 employees, including engineers, technicians, and administrative staff, the company is committed to talent development. Staff undergo regular training in areas like safety, quality protocols, and equipment usage. Internship programs, such as the one I participated in, demonstrate the organization's dedication to real-world skill-building and knowledge transfer.

Market & Clientele

Star Engineers serves a broad array of clients, supplying both Tier-1 and Tier-2 automotive vendors, as well as businesses in telecommunications, industrial automation, and consumer electronics. Their solutions frequently meet **ISO/TS 16949** and **ISO 9001** quality benchmarks, ensuring consistency and performance excellence.

Compensation & Reviews

Employees at Star Engineers report **competitive and timely salaries**, complemented by health benefits, performance bonuses, and career development opportunities. Overall employee feedback highlights an inclusive and supportive work environment, with strong managerial backing and a culture that values productivity equally with well-being.

CHAPTER 3: INTERNSHIP EXPERIENCE

As part of the final-year curriculum of Electronics and Telecommunication Engineering, I completed a 3-month internship at **Star Engineers India Pvt. Ltd., Chakan, Pune**, from **17th March 2025 to 17th June 2025**.

Iwas assigned to the **MI Department (Production Section)** as an **ICT Intern**, where I had the opportunity to engage in various hands-on activities related to production and testing.
My primary responsibilities included **testing PCB (Printed Circuit Board) components**, using in-circuit testing fixtures and **Kyoritsu Software**, under the guidance of experienced engineers.

Ialso assisted in **production documentation**, maintaining logs of test results, daily observations, and ensuring that data entry followed ISO-compliant formats.
Additionally, I provided **production support**, collaborating with team members to streamline assembly and inspection processes.

Working under **Mr. Suryakant Jadhav (Supervisor)** and mentors **Mr. Rahul Tarale** and **Mr. Datta Chavan**, I received technical guidance and practical exposure to real-time manufacturing challenges.
Through this internship, I gained valuable skills in **optical and visual inspection of PCBs**, which helped sharpen my attention to detail and technical accuracy.

Ilearned to handle **in-circuit testing equipment**, troubleshoot minor component faults, and interpret test results effectively.
I became proficient with**Kyoritsu Software**, which is used for analyzing electronic components in production environments.

In terms of soft skills, I significantly improved my **communication**, **teamwork**, and **coordination abilities** while working in a collaborative industrial environment.
The experience also enhanced my understanding of **productionmanagement**, **plan execution**, and **quality control** processes used in the electronics manufacturing sector.

One of the most insightful aspects was observing the **integration of automation** in testing and quality assurance activities.
The work culture at Star Engineers was professional and supportive, and I was encouraged to ask questions and take initiative.

Daily interactions with experienced engineers taught me the value of **discipline**, **safety awareness**, and **technical documentation** in a high-volume production setup.

Ifaced initial challenges in adapting to industrial workflows and using technical software, but with training and mentorship, I overcame them successfully.

Ialso observed how production data is monitored and decisions are made to optimize throughput without compromising quality.
Overall, the internship bridged the gap between academic learning and practical application in a real-world industrial context.

Itgave me confidence in my technical capabilities and helped me align my academic knowledge with industry expectations.
This experience was instrumental in preparing me for future professional roles in electronics manufacturing or embedded systems industries.

CHAPTER 4: INTERNSHIP OBJECTIVES

The main purpose of undertaking this internship at **Star Engineers India Pvt. Ltd.** was to gain practical exposure to industrial electronics and to enhance academic knowledge with real-world experience. As a final-year Electronics & Telecommunication Engineering student, this internship served as a critical bridge between classroom learning and professional practice.

1. To gain hands-on experience in electronic component manufacturing and testing.
2. To understand the working structure of a real-world production department (MI Department).
3. To learn the process of **in-circuit testing (ICT)** and fault detection in PCBs.
4. To acquire skills in **optical and visual inspection** of printed circuit boards.
5. To understand the use of industry tools like **Kyoritsu Software** and in-circuit fixtures.
6. To observe how documentation and production planning are managed in a manufacturing unit.
7. To become familiar with **quality assurance** standards in the electronics industry.
8. To understand the daily workflow and responsibilities of production support engineers.
9. To identify the **safety protocols and hazard control measures** followed in electronics manufacturing.
10. To enhance **problem-solving abilities** through real-time troubleshooting.
11. To practice **collaborative teamwork** and effective communication in an industrial setup.
12. To learn how data from testing equipment is recorded, analyzed, and reported.
13. To experience how production lines are set up, operated, and maintained.
14. To develop discipline, time management, and responsibility in a professional environment.
15. To improve the ability to adapt to industrial tools, software, and working methods.
16. To understand the **hierarchical structure** and workflow of departments in an electronics company.
17. To learn how **inter-departmental coordination** affects overall production efficiency.
18. To observe how customer quality expectations are translated into production goals.
19. To understand the lifecycle of an electronic component from design to final testing.
20. To gain exposure to **maintenance and calibration** of testing equipment.
21. To improve documentation skills related to test reports and observations.
22. To understand inventory tracking and how component shortages affect production.
23. To observe the importance of **lean manufacturing practices** in electronics assembly.
24. To get a first-hand understanding of **industry expectations from engineering graduates**.

CHAPTER 5: DEPARTMENT OVERVIEW

MI Department – Production Section:

The **MI Department** at **Star Engineers India Pvt. Ltd.**—also known as the **Production Section**—is a crucial operational unit responsible for the **assembly, testing, inspection, and production support** of electronic components. "MI" stands for **Manufacturing and Inspection**, and the department ensures that products meet stringent industry standards before reaching the client.

This department plays a pivotal role in executing the production plan, maintaining quality assurance, and supporting process optimization across the manufacturing floor. It acts as a bridge between design, testing, and final assembly, ensuring that each unit is built to exact specifications.

The MI Department is equipped with specialized tools and technologies such as **in-circuit testing (ICT) fixtures**, **Kyoritsu Software**, **digital inspection tools**, and **optical inspection stations**. It handles high-precision testing of **printed circuit boards (PCBs)**, detecting errors or defects early in the production cycle.

As an ICT Intern in this department, I was actively involved in key functions such as:

- Testing and inspecting assembled PCBs
- Documenting production results and test reports
- Supporting production staff in component preparation and system validation

The department emphasizes **accuracy, consistency, and safety** in all its operations. Each activity is monitored closely by engineers and quality officers to ensure compliance with ISO and client-specific standards.

Overall, the MI Department is an integral part of Star Engineers' commitment to **reliability, quality control**, and **continuous improvement** in the electronics manufacturing process.

CHAPTER 6: DETAILED WORK

RESPONSIBILITIES & TASKS

During my 3-month internship in the MI Department (Production Section) at Star Engineers India Pvt. Ltd., I was assigned various responsibilities that contributed to my technical learning and industrial exposure. My tasks were structured around production support, PCB testing, and documentation. Below is a breakdown of the key responsibilities handled during the internship:

1. PCB Testing (In-Circuit Testing – ICT)

- Operated in-circuit fixtures to test assembled PCBs for electrical and functional accuracy.
- Used Kyoritsu Software to run automated tests and interpret diagnostic results.
- Identified defects such as component misplacement, soldering faults, and continuity errors.
- Performed both powered and unpowered tests as per standard procedures.

2. Visual and Optical Inspection

- Conducted visual inspection of PCBs for alignment, color codes, and solder joints.
- Used magnifiers and optical aids to check component orientation and placement.
- Assisted in first piece inspection (FPI) before mass production to verify initial setup.

3. Production Documentation

- Maintained daily records of tested PCBs and noted error codes from ICT results.
- Helped prepare production reports, component traceability sheets, and rework logs.
- Ensured documentation was aligned with company protocols and audit requirements.

4. Production Support

- Assisted production operators in loading/unloading components on test jigs.
- Supported engineers in preparing boards for retesting after rework.
- Helped with organizing tested and faulty PCBs in designated storage areas.

5. Maintenance and Calibration Checks

- Participated in basic fixture cleaning and assisted in periodic calibration routines.
- Reported unusual fixture behavior or software errors to department supervisors.

6. Team Coordination and Learning

- Attended team briefings on daily goals and production plans.
- Learned communication etiquette for reporting issues and requesting support.
- Collaborated with senior engineers, interns, and technicians for task completion.

These responsibilities helped me understand the structured workflow in electronics manufacturing and the role of inspection and documentation in ensuring product quality. The hands-on experience strengthened my technical foundation and prepared me for future professional roles in the electronics and telecommunication domain.

CHAPTER 7: TOOLS & SOFTWARE USED

During my internship at **Star Engineers India Pvt. Ltd.**, I worked with various tools, equipment, and software essential to the **production and testing** of electronic components. These tools played a critical role in ensuring the accuracy, reliability, and quality of printed circuit boards (PCBs) and helped me gain hands-on technical experience.

🔧 1. Kyoritsu Software

- A specialized **testing anddiagnosticsoftware** used for in-circuit testing (ICT) of PCBs. Used to automate testing sequences, generate real-time results, and identify defects such as short circuits, open connections, or incorrect component values.
- Enabled logging and storing of test results for further documentation and analysis.

2. Desktop Computers & Workstations

- Used for running Kyoritsu Software, maintaining production reports, and accessing documentation.
- Assisted in preparing test logs, daily output summaries, and component tracking sheets.
- Also used for basic data entry, fixture settings, and reporting to supervisors.

🔩 3. In-Circuit Testing Fixtures

- Physical test benches with **spring-loadedtestprobes** used to make contact with PCB test points.
- Helped perform electrical and continuity tests on assembled boards without damaging components.
- Essential for verifying component installation, polarity, and soldering accuracy.

🔍 4. Optical Inspection Tools

- Included magnifying lenses and inspection lights to assist in **visual examination** of PCBs. Used to identify visible defects such as poor solder joints, misaligned components, or incorrect placements.
- Helped improve visual inspection accuracy, especially in high-density boards.

5. Basic Hand Tools

- Used toolslike **screwdrivers, tweezers, ESD-safe gloves**,and anti-static mats during testing and handling.
- Ensured proper handling of sensitive components while preventing electrostatic discharge (ESD) damage.

These tools and software formed the foundation of the testing and quality assurance process in the MI Department. My interaction with them provided valuable practical experience, enhanced my diagnostic skills, and familiarized me with industrial electronics testing procedures.

CHAPTER 8: SKILLS LEARNED

My 3-month internship at **Star Engineers India Pvt. Ltd.**, Chakan, Pune, offered a valuable opportunity to develop both **technical** and **professional** skills relevant to the electronics and telecommunication industry. These skills were gained through hands-on tasks, observation, guidance from mentors, and active participation in the production workflow.

🔧 1. Optical & Visual Inspection

- Gained proficiency in visually examining PCBs for soldering faults, component misalignment, missing parts, and physical defects.
- Learned to identify common manufacturing errors using inspection lamps and magnifiers.

🛠️ 2. In-Circuit Testing Knowledge

- Developed working knowledge of **ICT(In-Circuit Testing)** principles and procedures.
- Learned how to operate testing fixtures and interpret results from **Kyoritsu Software** to detect short circuits, open connections, and incorrect components.

🗂️ 3. Production Documentation

- Acquired skills in documenting test results, maintaining fault logs, and preparing production reports.
- Understood the importance of **traceability and record-keeping** in electronics manufacturing.

🗂️ 4. Production and Plan Management

- Observed how production lines are scheduled, monitored, and optimized for efficiency.
- Gained awareness of **workflow planning**, task assignment, and output tracking in a manufacturing environment.

5. Problem Identification & Resolution

- Developed analytical thinking to detect faults, understand their root causes, and support the resolution process.
- Worked closely with senior engineers to learn troubleshooting techniques for faulty boards.

6.Safety and Handling Practices

- Understood theuseof **ESD precautions**,safe handling of components, and personal protective equipment (PPE).
- Learned the importance of following industrial **safety standards and compliance protocols**.

7. Communication and Collaboration

- Improved verbal and written communication by interacting with supervisors, reporting issues, and participating in daily briefings.
- Collaborated with other interns, technicians, and department staff effectively in team-based activities.

These skills have not only enhanced my technical abilities but also prepared me for future roles in the electronics industry by giving me a strong foundation in production operations, quality assurance, and professional conduct.

CHAPTER 9: CHALLENGES FACED

During my internship at **Star Engineers India Pvt. Ltd.**, I encountered several challenges that tested my adaptability, technical understanding, and communication skills. Overcoming these hurdles significantly contributed to my learning experience and personal growth.

⚙️ 1. Adapting to the Industrial Environment

- Transitioning from an academic setup to a fast-paced production environment was initially challenging.
- Understanding the workflow, discipline, and coordination required in a professional manufacturing setting took time and observation.

🛠️ 2. Limited Initial Knowledge of ICT Tools

- I had no prior hands-on experience with **in-circuit testing fixtures** or **Kyoritsu Software**.
- Learning how to operate the testing setup, interpret software results, and respond to test failures required extra effort during the initial phase.

📑 3. Understanding Production Documentation

- Interpreting fault logs, production records, and component traceability documents was initially confusing due to unfamiliar formats and terminology.
- With guidance from mentors, I gradually learned how to maintain and update these records accurately.

🧩 4. Handling Repetitive Tasks with Precision

- Performing routine inspection and testing tasks required **sustained focus** and **attention to detail**.
- Maintaining consistency without compromising accuracy was a skill I had to consciously develop over time.

5. Managing Errors and Rework

- Sometimes, tested boards failed due to undetected faults, requiring rework and retesting.
- Reporting and escalating issues without delay, and assisting in rectification under time constraints, was a pressure-driven learning experience.

🧠 **6. Technical Communication**

- Explaining problems clearly to engineers and technicians required the use of correct technical terms and effective communication.
- I learned how to **document issues** properly and convey them clearly in discussions and daily reports.

7. Time Management and Task Prioritization

- Balancing multiple tasks such as testing, documentation, and reporting within tight timelines was challenging.
- I had to learn how to prioritize tasks based on urgency, accuracy, and production targets.

Despite these challenges, the support and mentorship from experienced staff helped me overcome them and convert these difficulties into valuable learning moments.

CHAPTER 10: INTERNSHIP IMAGE GALLERY

The following images were captured during my internship in the MI Department (Production Section) at Star Engineers India Pvt. Ltd., Chakan, Pune. They showcase the tools, equipment, and production environment I worked in during the 3-month period.

📸1. In-Circuit Test Fixtures

A rack of multiple in-circuit testing fixtures used for testing PCBs with various pin layouts. These were regularly used during my internship to test the functionality and connection accuracy of electronic boards.

Fig 10.1: ICT - Fixtures

🏭 2. Production Line Setup

This image shows the main assembly line in the MI department. Production status boards and SOP (Standard Operating Procedures) are displayed above the line for reference and quality compliance.

Fig 10.2: Production Line

⬛3. In-Circuit Testing with Kyoritsu Software

A working ICT machine where PCBs are tested using Kyoritsu Software. The green “PASS” status indicates that the boards passed all electrical and component-level checks.

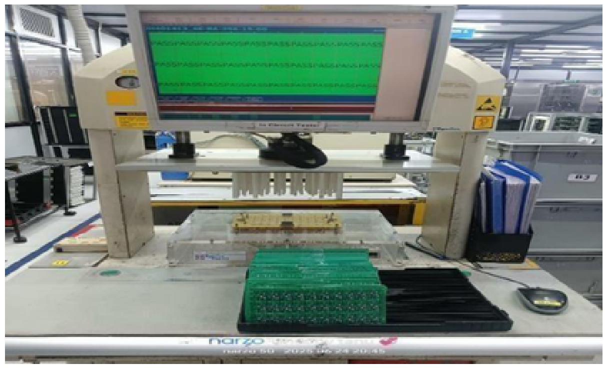

Fig 10.3: ICT Machine

Q •4. AOI – Automatic Optical Inspection Machine

This is the Omron RNS II-pt AOI machine, used to visually inspect PCBs for defects. It provides high-speed inspection to ensure component placement and soldering quality.

Fig 10.4: AOI Machine

🔧 5. Component Storage Bins

Storage drawers labelled with various resistors and capacitors used during the production process.

Iused these components during fixture setup and rework support tasks.

Fig 10.5: Component Storage Bins

These images help reflect the real-time industry exposure, tools I used, and the standard operating environment of Star Engineers. They also serve as a visual testament to the hands-on learning I gained
during this internship.

CHAPTER 11: CONCLUSION

My 3-month internship at **Star Engineers India Pvt. Ltd., Chakan, Pune**, has been a highly enriching and transformative experience. Being part of the **MI Department (Production Section)** allowed me to gain **real-world exposure** to industrial practices in electronics manufacturing, especially in the areas of **PCB testing, production documentation**, and **quality control**.

Throughout the internship, I was able to bridge the gap between classroom learning and actual industry application. I had the opportunity to work hands-on with advanced equipment such as **in-circuit testing (ICT) fixtures**, **Kyoritsu Software**, and **optical inspection tools**, which helped me develop practical skills and technical confidence.

Working under the guidance of experienced professionals like **Mr. Suryakant Jadhav**, **Mr. Rahul Tarale**, and **Mr. Datta Chavan** not only improved my understanding of the production process but also taught me the importance of teamwork, discipline, and communication in a corporate environment.

The knowledge I gained in **visual inspection, safety assessment, problem-solving**, and **production planning** has significantly strengthened my foundation as an aspiring engineer. I also learned how to effectively document production data, follow quality protocols, and maintain accuracy under tight timelines.

Most importantly, this internship has shaped my **professional mindset**, improved my time management skills, and prepared me to handle real-world challenges in the electronics and telecommunication field. In conclusion, the internship has been an essential step in my career development, offering me a deeper insight into industrial operations and setting a clear direction for my future goals. I am grateful to Star Engineers India Pvt. Ltd. and my college for this opportunity, which has truly enhanced my academic and professional journey.

CHAPTER 12: REFERENCES

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   Omron Industrial Automation

5. ISO 9001 & ISO/TS 16949 Standards
   International Organization for Standardization
   https://www.iso.org/

6. Production and Quality Documentation Procedures
   Internal SOPs and departmental documentation from Star Engineers India Pvt. Ltd. (accessed during internship)

7. Personal Photographs taken during internship at Star Engineers India Pvt. Ltd., Chakan – used with permission for academic submission.